



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No : ECE F311
Course Title : Communication Systems
Instructor-in-Charge : Prof. Syed Mohammad Zafaruddin
Instructors : Prof. Syed M Zafaruddin (L1) and
Prof. Shubha Sharma (L2)

Tutorial Instructor(s): Prof. Syed M Zafaruddin (T1), Prof. Rahul Singhal (T2), Prof. Shubha Sharma (T3), Prof. Monolina Dutta (T4), Prof. Ritesh Vyas (T5)

Practical Instructor (s): Ananya Garg (P1, P5), Akhil Misra (P1, P5), Rayees Ahmad Wani (P2, P6), Arvind Kumar (P2, P3), Tejas Sharma (P3, P4), Aishwarya B. Mudennavar (P4), Vipin Kr Saraogi (P6)

Course Description:

Analysis and design of communication systems; analog and digital modulation and demodulation, frequency conversion, multiplexing, noise and distortion; spectral and signal-to-noise ratio analysis, probability of error in digital systems, spread spectrum. Introduction to the basic principles of the design and analysis of modern digital communication systems. Topics include source coding, channel coding, baseband and passband modulation techniques, receiver design, and channel equalization

1. Scope and Objective of the Course:

This course intends to cover the basic understanding of functionalities of various block-sets involved in communication system. Course starts with an overview of communication systems, communication process, generic model and block diagram description. The course also reviews important concepts of signal and systems, and random variables which are useful for understanding and analyzing communication systems. The course covers both analog and digital communication systems. The course also covers information theoretic approach to deal with the random additive noise.

The course deals with analog and digital modulation schemes in a deterministic way as well as analyzes the effect of random noise on the performance of communication systems. The course introduces performance metrics such as SNR and BER and performance limits of communications for both analog (using Nyquist rate) and digital (using Shannon theorem). The course has a lab component which includes simulation experiments for analog and digital communication systems using MATLAB and Python software.

The objective of the course is to equip students with fundamental understanding to model, design, and analyze analog and digital communication systems for various practical scenarios. This course will serve as a basis to learn advanced level courses on communications and develop physical layer algorithms. Students will be given assignments on communication system modeling using MATLAB and Python. Students are expected to have an understanding of Signals and Systems, Electronic Devices and Circuits, Mathematics, and basics of Electromagnetic Field Theory.

2. Text Books:

T1: B.P. Lathi and Zhi Ding, “*Modern Digital and Analog Communication Systems*”, 4th Edition, Oxford University Press, 2017, South Asia Edition (Adapted by H.M. Gupta)

T2: Simon Haykin & Michael Moher, “*Communication Systems*”, 5th Edition, WSE, John Wiley & Sons, 2010.

3. Reference Books:

- R1: John G. Proakis, Masoud Salehi, “*Communication Systems Engineering*”, 2nd Edition, Pearson, 2002.
- R2: Simon Haykin, “*Communication Systems*”, 4th Edition, John Wiley & Sons, 2001.



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- R3: A. Papoulis, S. Unnikrishna Pillai, “*Probability, random variables, and stochastic processes*”, 4th Edition, TMH, 2002.

4. Course Plan:

Module No.	Lecture Session	Reference	Learning Objectives
(I): Introduction to Communication Systems and Bandwidth (2 Lectures)	Lectures 1–2: Overview of the course; taxonomy of communication systems; block-level description of an end-to-end communication system; baseband and passband transmission; brief review of the Fourier transform and signal spectrum; signal bandwidth, channel bandwidth and bandwidth requirements in communication systems.	T1: chapters 1, 2	Describe the functional blocks of an end-to-end communication system; distinguish between baseband and passband transmission; interpret communication signals in the time and frequency domains; and relate signal bandwidth to channel bandwidth.
(II): Analog modulation and Demodulation (12 Lectures)	Lectures 3–10: Amplitude-modulation techniques: DSB-SC, AM with carrier, SSB-SC, VSB; angle-modulation techniques: frequency modulation and phase modulation; bandwidth and power requirements; generation of AM and FM signals; Armstrong method of FM generation; and frequency-division multiplexing.	T1: chapters 3,4	Explain frequency translation and the principles of amplitude and angle modulation; compare analog modulation schemes in terms of bandwidth and power requirements; describe methods for generating AM, FM and PM signals; and explain frequency-division multiplexing.
	Lectures 11–14: Synchronous detection; envelope detection; FM and PM demodulation; phase-locked-loop-based demodulation; and superheterodyne receivers.	T1: chapter 3,4	Explain synchronous and envelope detection; describe FM- and PM-demodulation techniques, including PLL-based demodulation; and explain the architecture and operation of a superheterodyne receiver.
(III) Communication Channels and Noise (10 Lectures)	Lectures 15-19: Communication media and channel models; signal transmission over communication channels; conditions for distortionless transmission; amplitude and phase distortion; random variables and random processes; statistical averages; autocorrelation and power spectral density; Gaussian processes; bandpass random processes; sources and nature of noise; additive white Gaussian noise; and noise figure.	T1: chapter 2,6,7	Explain communication-media and channel models; identify amplitude and phase distortion; apply random-variable and random-process concepts in communication-system analysis; interpret autocorrelation and power spectral density; and explain Gaussian noise, bandpass noise, AWGN and noise figure.



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	Lectures 20-24: Analog demodulation in the presence of additive noise; output signal-to-noise ratio; noise performance of AM, DSB-SC, SSB-SC, FM and PM receivers; threshold effect in angle modulation; and comparison of analog modulation systems under additive noise.	T1: chapter 7	Analyse the effect of additive noise on analog demodulators; determine the output signal-to-noise ratio; compare the noise performance of amplitude- and angle-modulation systems; and explain the threshold effect in angle modulation.
(IV): Digital Signal Representation, Modulation and Detection (12 Lectures)	Lectures 25–29: Pulse modulation; quantisation and encoding; pulse-code modulation; delta modulation; baseband line-coding schemes, including NRZ, polar and Manchester signalling; power spectral density of line-coded signals; intersymbol interference; eye diagrams; Nyquist criterion for distortionless transmission; pulse shaping; and waveform representation of binary and M-ary passband modulation schemes, including ASK, PSK, QAM and FSK.	T1: chapter 5, 8, 11	Explain digital representation using PCM and delta modulation; compare common line-coding schemes and their power spectra; analyse intersymbol interference using eye diagrams; apply the Nyquist criterion and pulse-shaping concepts; and interpret binary and M-ary digital-modulation waveforms.
	Lectures 30–36: Digital demodulation in the presence of additive noise; detection of digital signals in noise; optimum-threshold detection; probability of error and bit-error probability; matched filters and optimum receivers; and coherent and noncoherent detection of digital carrier-modulation schemes.	T1: chapter 8, 9	Analyse the detection of digital signals in additive noise; determine optimum decision thresholds and bit-error probability; explain matched-filter and optimum-receiver principles; and compare coherent and noncoherent detection techniques.
(V): Information Theory and Channel Coding (6 Lectures)	Lectures 37-40 Measure of information; entropy; source coding; Shannon–Fano coding; Huffman coding; data compression; mutual information; channel capacity; capacity of the AWGN channel; Shannon’s theorem and fundamental communication limits; and digital-transceiver design considerations.	T1: chapter 12	Explain the measures of information and entropy; apply source-coding and data-compression techniques; interpret mutual information and channel capacity; explain Shannon’s theorem and the capacity of an AWGN channel; and identify fundamental trade-offs in digital-transceiver design.
	Lectures 41-42 Error Detection and Correction Codes: Hamming codes; linear block codes; and an introduction to cyclic codes	T1: chapter 13	Explain the principles of error detection and correction; apply basic Hamming and linear block codes; and describe the fundamental concept of cyclic codes.



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5. Practical:

The weekly lab series is structured to guide students through the progressive development of a modular, **end-to-end communication system simulator**. Implemented in **MATLAB**, the simulator emphasizes the use of data structures and object-oriented programming to model each communication block clearly and flexibly. Throughout the course, students will also perform standalone validation of individual modules using MATLAB and Emona hardware kits, with the possibility of integrating the complete system with USRP-based Software-Defined Radios (SDRs) in the final stages.

The core objective is to bridge theoretical understanding with simulation, visualization, and hands-on experimentation in a structured and modular way.

Weekly Lab Schedule:

Week 1	Introduction to information source modeling: analog and digital signal, with data structure implementation and time/frequency analysis: class InfSource
Week 2	Design of transmitter block with emphasis on convolution-based filtering for bandwidth-limited signals class Tx
Week 3	Implementation of bandlimited channels (wireline and wireless) and noise: class channel
Week 4	Updating the Tx with passband analog modulation schemes Develop various receiver structure with performance metrics SNR and RSSI: class Rx.
Week 5	One Lab demonstration on Analog modulation using Emona hardware kits
Week 6	Real-time FM/AM E2E transmission/reception for audio signal of a 4-minute duration.
Week 7	Updating the Tx and Rx class with digital baseband transmission: implementation of ADC/DAC and Nyquist Signalling
Week 8	Design the matched filter receiver and analyze the BER: update class Rx
Week 9	One Lab demonstration on Digital modulation using Emona hardware kits
Week 10	Signal constellation based M-ary digital modulation scheme: Update classes Tx and Rx and real-time transmission of one text sentence.
Week 11	Channel coding and decoding: Update classes Tx and Rx with real time transmissions of one page text.
Week 12	Introduction to the SDR: Integrating the E2E with SDR

Laboratory Project: Demonstration and evaluation of a complete end-to-end communication-system simulator supporting real-time audio and text transmission, together with the investigation and implementation of a defined research-oriented problem. Assessment will be based on system functionality, technical innovation, performance analysis, demonstration, report and viva.



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6. Evaluation Scheme:

Component	Duration	Marks (300)	Date & Time	Nature of component
Quizzes	20 min	15% (45)	Quizzes (Best 5 of 7)	Close Book
Weekly Labs	2 hours	16.67% (50)	Continuous	Open/Close Book
Lab Project	-	8.33% (25)	Continuous	Demonstration, report and viva
Mid-Semester Test	As per AUGSD	25% (75)	As per AUGSD	Close Book
Comprehensive Exam	As per AUGSD	35% (120)	As per AUGSD	Open Book

7. Notices: All notices will be displayed on Learning Management System (LMS-NALANDA).

8. Make-up Policy: No make-up will be given for Quizzes. For other components, make up will be given only on genuine reasons. Applications for makeup should be given in advance and prior permission should be obtained for scheduled tests.

Instructor-in-charge
Course No. ECE/ EEE F311